Temperature, VPD, and Global Crop Yields — Replication Materials

Tosun, A.U. and Karaman, F. (2025)

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Overview

This repository contains the code and documentation needed to replicate all analyses in the paper. The study examines the effect of flowering-period maximum temperature (TMX) and vapor pressure deficit (VPD) on crop yields across eight staple crops using a global country-year panel dataset covering 1961–2024. A key contribution is the mechanistic separation of TMX-based and VPD-based yield effects via a Clausius–Clapeyron decomposition, and the application of CMIP6 climate projections to estimate end-of-century yield risks under SSP2-4.5 and SSP5-8.5.

> **Note on code scope:** This repository includes all scripts used during the project, some of which generate exploratory figures and diagnostics not included in the final manuscript. Scripts are clearly named by step; publication-ready outputs are produced by `paper1_step11_tables_figures.R` and `paper1_ga_figures.R`.

> **Note on raw data:** No raw data files are included. All input data are publicly available from the sources listed below.

Data Sources

Source	Variable	Link
FAOSTAT	Crop yield, harvested area, production, irrigation data	https://www.fao.org/faostat
CRU TS4.09	Monthly max temperature, precipitation, vapour pressure	https://crudata.uea.ac.uk/cru/data/hrg
SPAM2020	Crop-specific harvested area weights	https://www.mapspam.info
World Bank WDI	GDP per capita (constant 2015 USD)	https://data.worldbank.org
GADM v4.1	Country boundaries	https://gadm.org
CMIP6 / WorldClim	Future temperature projections	https://worldclim.org/data/cmip6

Panel structure: Country × Year × Crop.
Coverage: 61–99 countries depending on crop, 1961–2024.

Key Variables

Primary variables (M1–M4 baseline models)

R variable	Description
`ln_verim`	Log crop yield — hg/ha (Turkish: *verim* = yield)
`TMX`	Flowering-period maximum temperature (°C), production-weighted
`PRE`	Precipitation during grain-filling period (mm)
`PRE_sq`	Squared precipitation — nonlinear precipitation response; significant concave relationship for rye only (see Appendix Table A.11)
`irr_ratio`	Irrigation intensity (ratio, 0–1) — equipped irrigation area / cropland area
`ln_gdp`	Log GDP per capita (constant 2015 USD)
`Country_ISO`	Country identifier — ISO3 code (used as country fixed effect)
`Yil`	Year (Turkish: *yıl* = year; used as year fixed effect)

Derived variables (VPD decomposition — Step 10)

R variable	Description
`VPD`	Vapor pressure deficit during flowering (kPa), derived from CRU vapour pressure and TMX via Clausius–Clapeyron: $e_s(T) = 0.6108 \times \exp[17.27T/(T+237.3)]$

Interaction terms (M5–M8 robustness specifications)

R variable	Description
`TMX × irr_ratio`	Temperature × irrigation interaction
`TMX × is_arid`	Temperature × aridity indicator (is_arid = 1 if annual PRE < 400 mm)
`TMX × ln_gdp`	Temperature × income interaction
`TMX × PRE`	Temperature × precipitation interaction

Nonlinearity tests (Appendix A.6 — see Appendix for full results)

R variable	Description
`TMX_sq`	Squared TMX — tested in Appendix Table A.10. Significant only for soybean (TMX* = 24.4°C) and sorghum (TMX* = 28.5°C); linear specification retained in main models for all eight crops.

Empirical Strategy

Two-way fixed effects (TWFE) panel model estimated separately for each crop using `fixest::feols`. Production-weighted estimates are the primary specification. VPD is derived from CRU vapour pressure and temperature data using the Clausius–Clapeyron relationship and translated into temperature-equivalent units to allow direct comparison with TMX-based projection losses.

Code Structure

The project is organized into two scripts.

`step00_datapreparation_paper1.R` — Data Preparation Pipeline

Builds the analysis-ready panel datasets from raw sources. Runs sequentially through 7 steps, each producing intermediate files (`pan1*` → `pan7*`).

Step	What it does	Output
Step 1	CRU TS4.09 NetCDF + SPAM2020 rasters → monthly climate panels	`pan1*`
Step 2	FAOSTAT yield + phenological windows → annual crop panels	`pan2*`
Step 3	Extra 4 countries (IRN, EGY, NGA, SDN) added via supplementary FAO data	`pan3*`
Step 3b	Egypt boundary patch via rnaturalearth	`pan3*` update
Step 4	FAOSTAT irrigation data joined	`pan4*`
Step 5	Cropland area + `irr_ratio` constructed	`pan5*`
Step 6	Aridity indicator (`is_arid`: annual PRE < 400 mm) added	`pan6*`
Step 7	GDP per capita (World Bank WDI) joined; backfill fix + duplicate removal	`pan7*`

Input files (all in `ana_yol`):

File	Description
`CRU_TS4.09_*.nc`	CRU TS4.09 monthly climate grids
`spam2020_V2r0_global_H_*_A.tif`	SPAM2020 harvested area rasters
`Global_AgriClimate_v5.xlsx`	Phenological windows (flowering, grain-filling months) per country × crop
`Production_Crops_Livestock_E_All_Data_(Normalized).csv`	FAOSTAT crop yield, area, production
`GLOBAL_ATMOSPHERIC_CO2_NOAA.csv`	Atmospheric CO_2 (NOAA)
`gdp_per_person.csv`	Fallback GDP (overridden by WDI in Step 7)
`FAOSTAT_data_en_3-5-2026.xls`	FAOSTAT irrigation data
`sudanvs.xls`	FAO data for IRN, EGY, NGA, SDN

Final outputs: `pan7wheat.xlsx` … `pan7sorghum.xlsx` (one file per crop)

Required packages: `terra`, `geodata`, `dplyr`, `readxl`, `writexl`, `data.table`, `tidyr`, `zoo`, `rnaturalearth`, `rnaturalearthdata`, `sf`, `WDI`

`step01_dataanalysis_paper1.R` — Data Analysis and Figures

Consolidates all analysis scripts into a single file. Reads `pan7*.xlsx` outputs from Step 00 and produces all models, figures, and tables for the manuscript and appendix.

Section	What it does	Output in manuscript?
Descriptive statistics	Summary plots and statistics	Partial
Correlation analysis	Correlation matrix and plots	No (exploratory)
Main TWFE models	Eight model variants (M1–M8), production- and area-weighted — saves `modeller_v4_objects.rds`	Yes
Precipitation robustness	PRE-squared analysis, optimal PRE estimates	Appendix
Coefficient summary	Forest plots, literature benchmark comparisons	Yes
Structural break & diagnostics	Chow tests, Pesaran CD tests, panel unit root	Yes
Leave-one-out robustness	Excluding USA, China, India, USSR bloc	Appendix
Driscoll-Kraay SE	Alternative inference, spatial autocorrelation	Appendix
Warming scenarios	Delta-T projections (+1°C to +4°C)	Yes
VPD decomposition	Clausius–Clapeyron VPD derivation, TMX vs VPD comparison — saves `adim10b_vpd_df.rds`	Yes
Publication figures & tables	Final manuscript figures, CMIP6 SSP2-4.5 / SSP5-8.5 projections	Yes
Graphical abstract	Figures B, C, D for graphical abstract	Yes

Required packages: `fixest`, `ggplot2`, `dplyr`, `tidyr`, `readxl`, `writexl`, `scales`, `MASS`, `plm`, `zoo`, `data.table`, `FAOSTAT`

> **Model specifications (M1–M8):** Full model equations, robustness tests, and diagnostic results are documented in Appendix A.0 of the manuscript.

> **Dependencies:** Steps 04–11 require `modeller_v4_objects.rds` saved by Step 03. Step 11 additionally requires `adim10b_vpd_df.rds` saved by Step 10.

How to Reproduce

1. Download all raw data from the sources listed above.
2. Set `ana_yol` at the top of each script to your local working directory.
3. Run `step00_datapreparation_paper1.R` first — produces `pan7*.xlsx` files for all eight crops.
4. Run `step01_dataanalysis_paper1.R` — reads `pan7*.xlsx` and produces all models, figures, and tables.

Simplified reproduction (recommended)

The analysis-ready panel datasets (`pan7wheat.xlsx`, `pan7barley.xlsx`, `pan7oats.xlsx`, `pan7rye.xlsx`, `pan7rice.xlsx`, `pan7maize.xlsx`, `pan7soybean.xlsx`, `pan7sorghum.xlsx`) are provided directly in this repository. Researchers who wish to replicate the statistical analyses and figures without re-running the full data preparation pipeline can do so by:

1. Setting `ana_yol` to their local working directory.
2. Running `step01_dataanalysis_paper1.R` only.

All results reported in the manuscript can be reproduced in this way.

Outputs

Type	Location	Format
Manuscript figures	`figures/labeled/` and `figures/unlabeled/`	PDF, 300 dpi PNG, 600 dpi PNG
Intermediate panels	`pan7wheat.xlsx` … `pan7sorghum.xlsx`	Excel
Model objects	`modeller_v4_objects.rds`	R binary
VPD data	`adim10b_vpd_df.rds`	R binary

Software and Package Versions

All analyses were conducted in **R version 4.5.2**.

Analytical packages (cited in main manuscript)

Package	Role	Version
`fixest`	TWFE panel estimation, clustered SE	0.13.2

Data access packages

Package	Role	Version
`FAOSTAT`	Automated FAOSTAT data download	2.4.0
`WDI`	World Bank WDI data download	2.7.9
`geodata`	Geographic and climate data download	0.6.6

Data processing packages

Package	Role	Version
`dplyr`	Data manipulation	1.2.0
`tidyr`	Data reshaping	1.3.2
`data.table`	Fast data processing	1.18.2
`zoo`	Time series operations	1.8.15
`readxl`	Excel file input	1.4.5
`writexl`	Excel file output	1.5.4
`plm`	Panel data utilities, Driscoll-Kraay SE	2.6.7
`MASS`	Robust regression utilities	7.3.65

Spatial packages

Package	Role	Version
`terra`	Raster processing (CRU, CMIP6 data)	1.8.93

Package	Role	Version
`sf`	Vector spatial data	1.0.24
`rnaturalearth`	Country boundary maps	1.2.0
`rnaturalearthdata`	Natural Earth dataset dependency	1.0.0

Visualization packages

Package	Role	Version
`ggplot2`	Figure generation	4.0.2
`scales`	Axis formatting in figures	1.4.0

Package Installation

```r
install.packages(c(
  "fixest", "ggplot2", "dplyr", "tidyr",
  "readxl", "writexl", "terra", "sf",
  "geodata", "rnaturalearth", "rnaturalearthdata",
  "scales", "MASS", "plm", "zoo",
  "data.table", "FAOSTAT", "WDI"
))
```

Citation

If you use this code or data pipeline, please cite:

> Tosun, A.U. and Karaman, F. (2025). Temperature, Vapor Pressure Deficit, and Global Crop Yields: Evidence from a Multi-Crop Panel. *Manuscript under review, Global Change Biology.*

For the `fixest` package:

> Bergé, L. (2018). Efficient estimation of maximum likelihood models with multiple fixed-effects: the R package FENmlm. *CREA Discussion Papers*, no. 13. https://lrberge.github.io/fixest/

For R itself:

R Core Team (2025). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/

Contact

Alper Umut Tosun
Yıldız Technical University, Istanbul, Turkey
umut.tosun@std.yildiz.edu.tr
ORCID: https://orcid.org/0000-0003-1167-8194

Prof. Dr. Filiz Karaman *(Corresponding author)*
Yıldız Technical University, Istanbul, Turkey
fkaraman@yildiz.edu.tr
ORCID: https://orcid.org/0000-0002-8491-674X